



# Engineering Mechanics

## Lecture 10: Wedges

By

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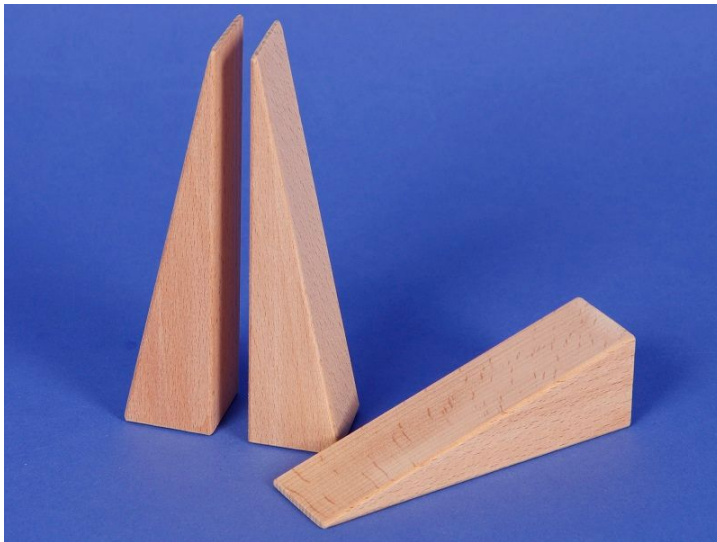
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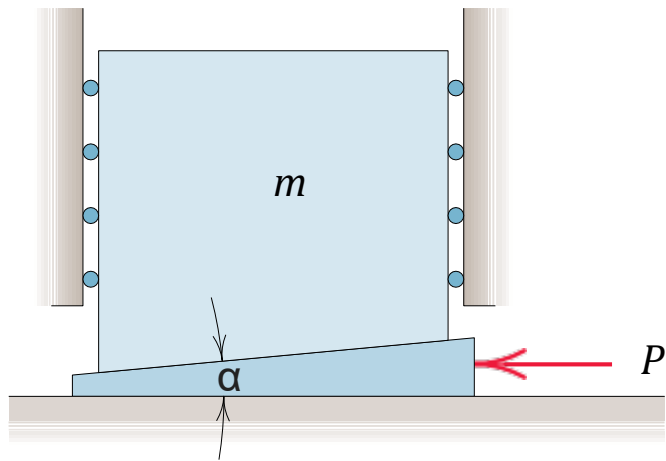
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# Wedge

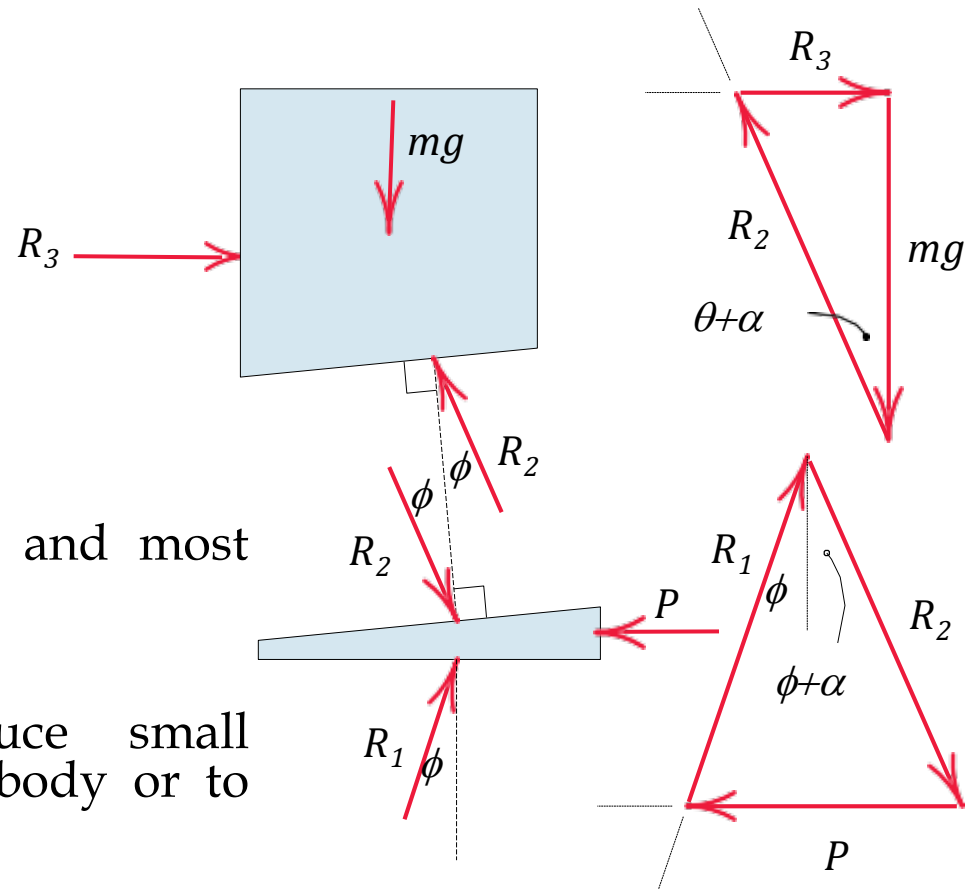
- A wedge is a triangular-shaped tool or object with a thin edge that tapers to a thicker end.
- It is used to split, lift, or secure objects by applying force, such as in cutting wood with an axe or lifting heavy objects.



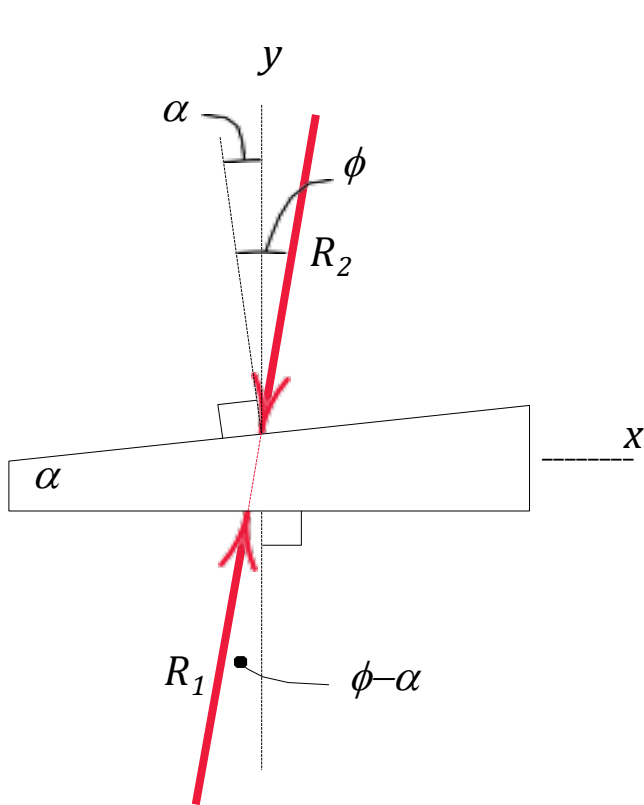
# Wedge: Raising Weight



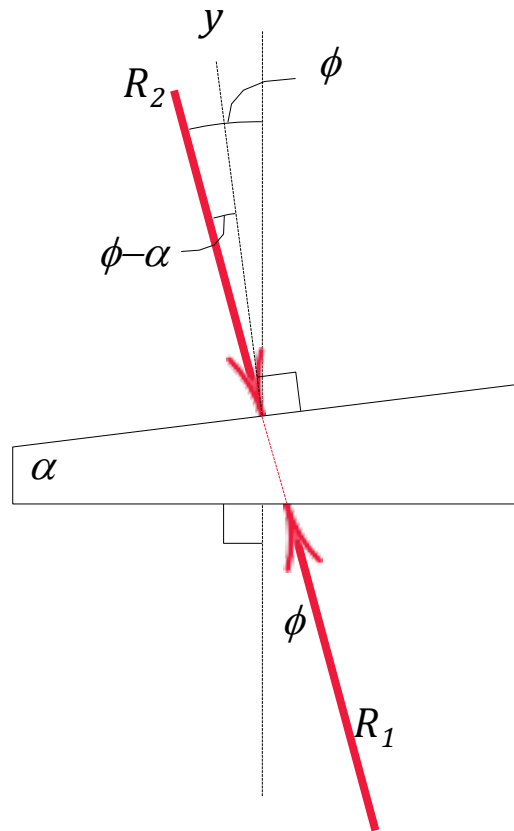
- A wedge is one of the simplest and most useful machines.
- A wedge is used to produce small adjustments in the position of a body or to apply large forces.
- Wedges largely depend on friction to function.



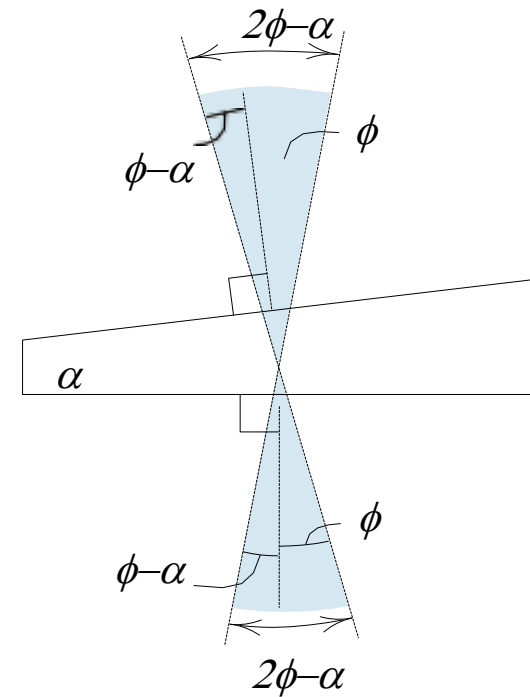
Force  $P = 0$  ( $R_1 = R_2$  for equilibrium)



Slipping impending  
at upper surface

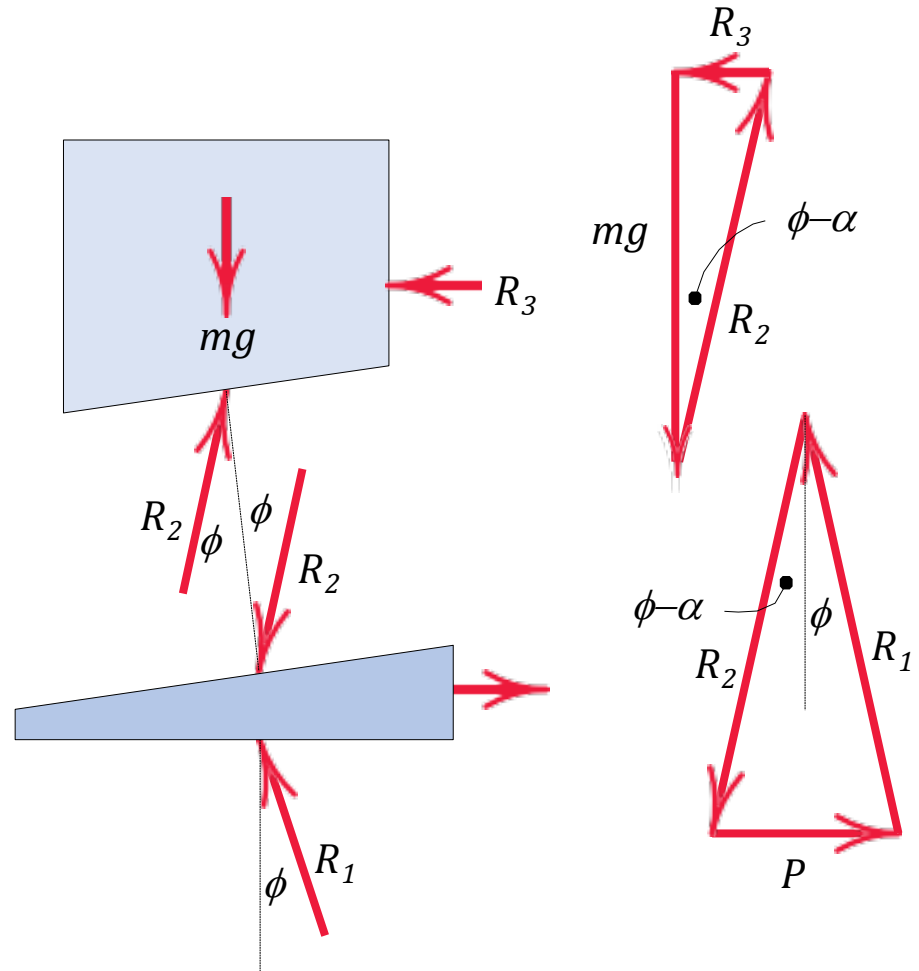
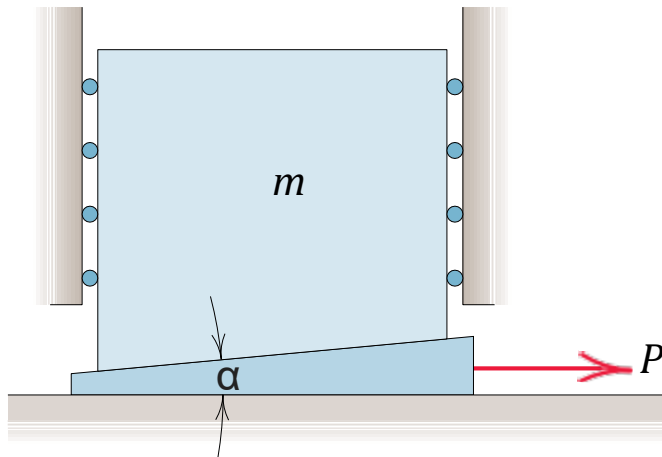


Slipping impending  
at lower surface

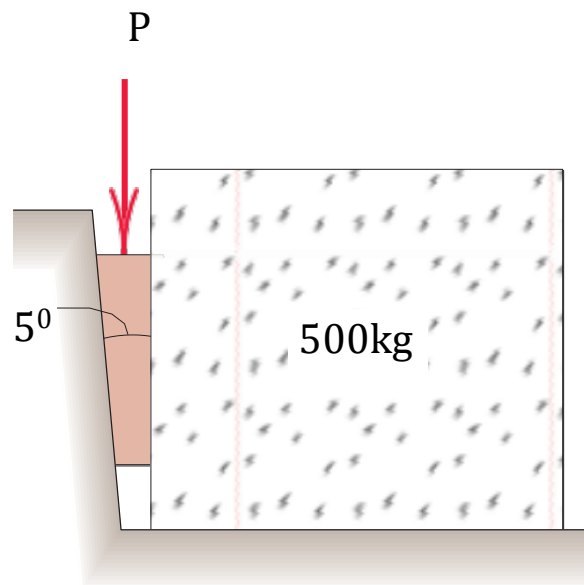


Range of  $R_1 = R_2$   
for no slip

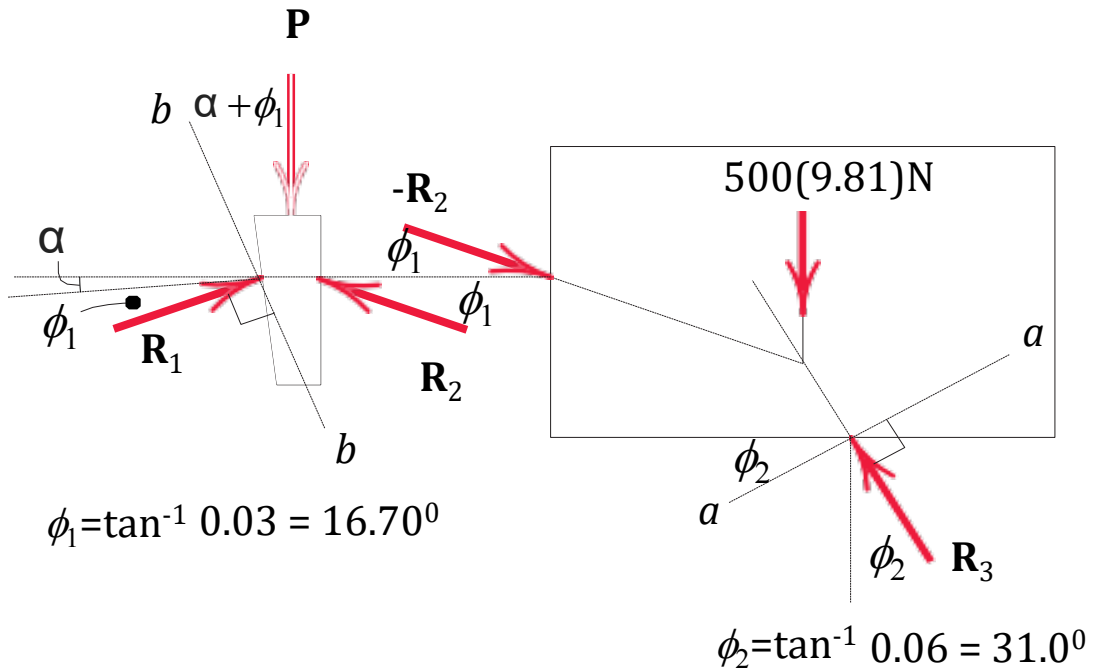
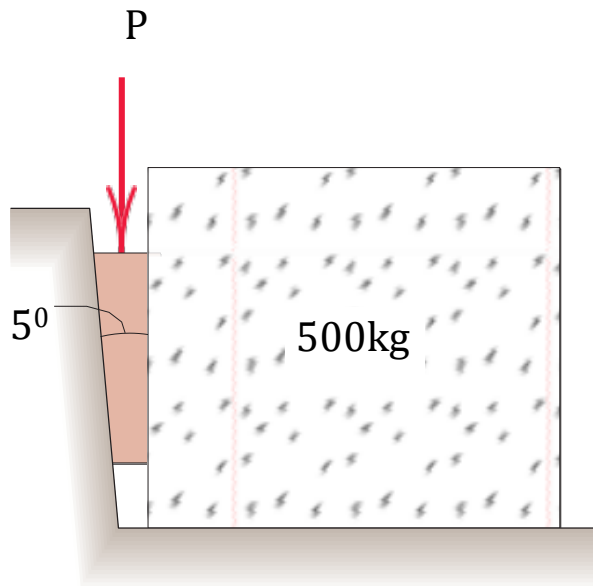
# Wedge: lowering weight



- **Example 2:** The horizontal position of the 500-kg rectangular block of concrete is adjusted by the  $5^\circ$  wedge under the action of the force  $P$ . If the coefficient of static friction for both wedge surfaces is 0.30 and if the coefficient of static friction between the block and the horizontal surface is 0.60, determine the least force  $P$  required to move the block.



# Algebraic solution



$$[\Sigma F_a = 0]$$

$$500(9.81) \sin 31.0^\circ - R_2 \cos 47.7^\circ = 0$$

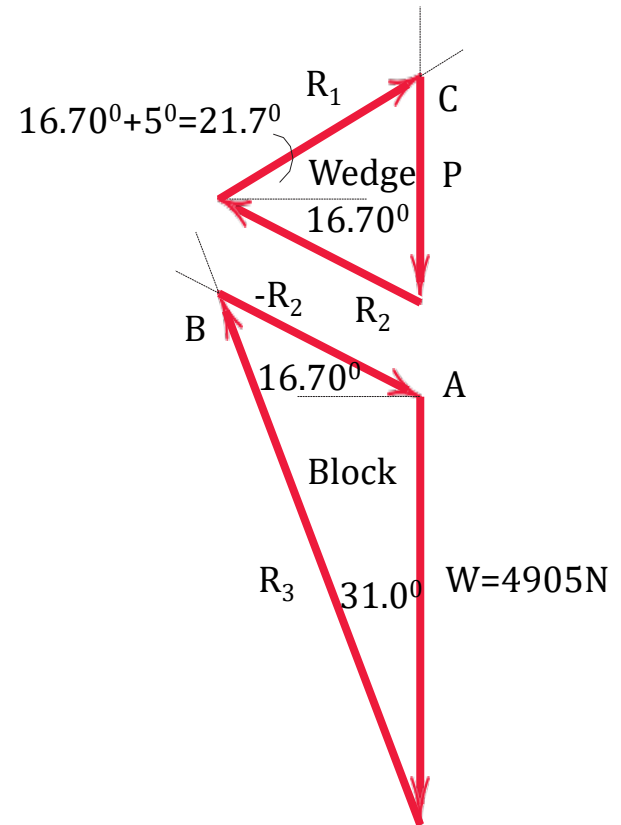
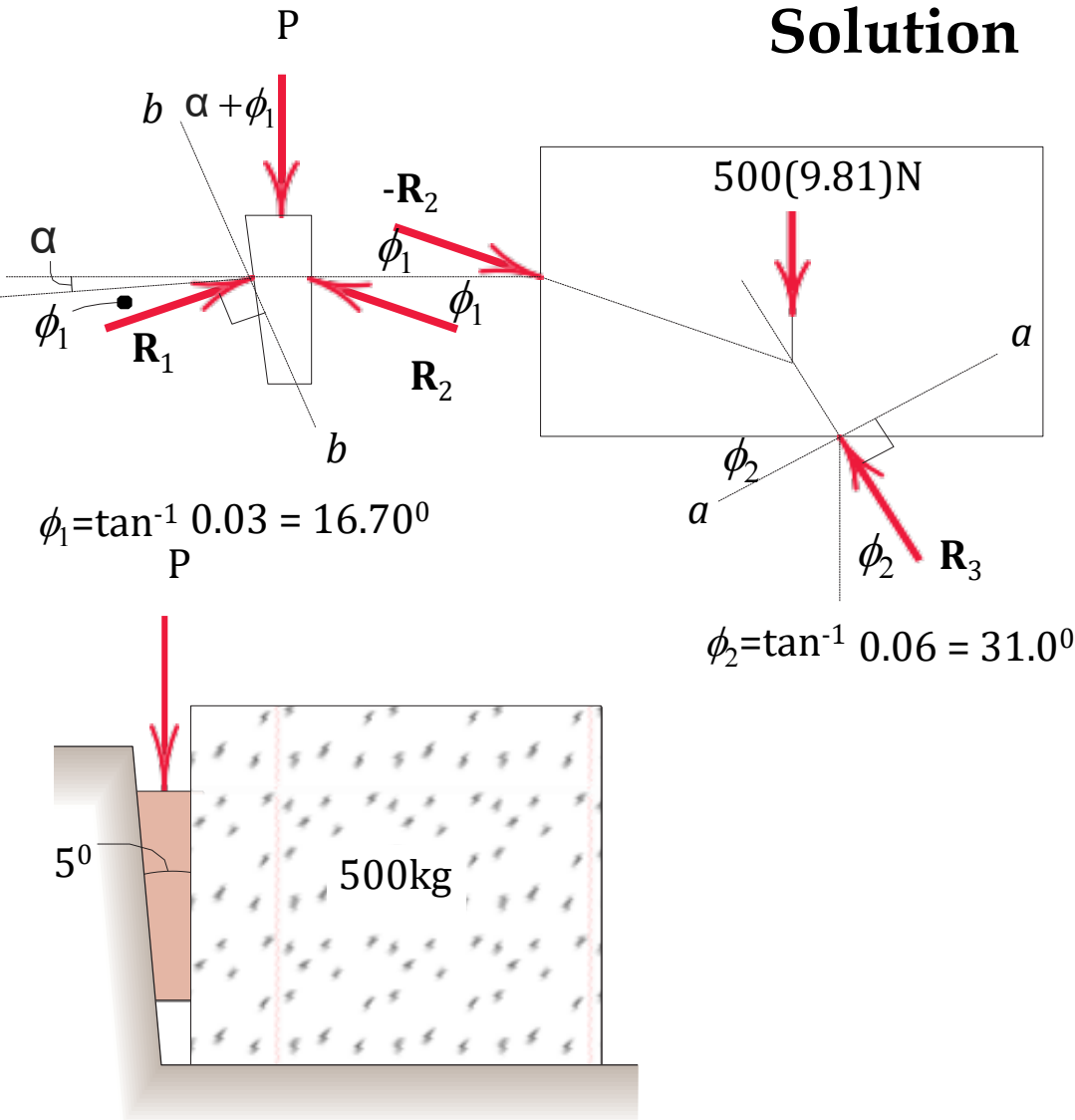
$$R_2 = 3750 \text{ N}$$

$$[\Sigma F_b = 0]$$

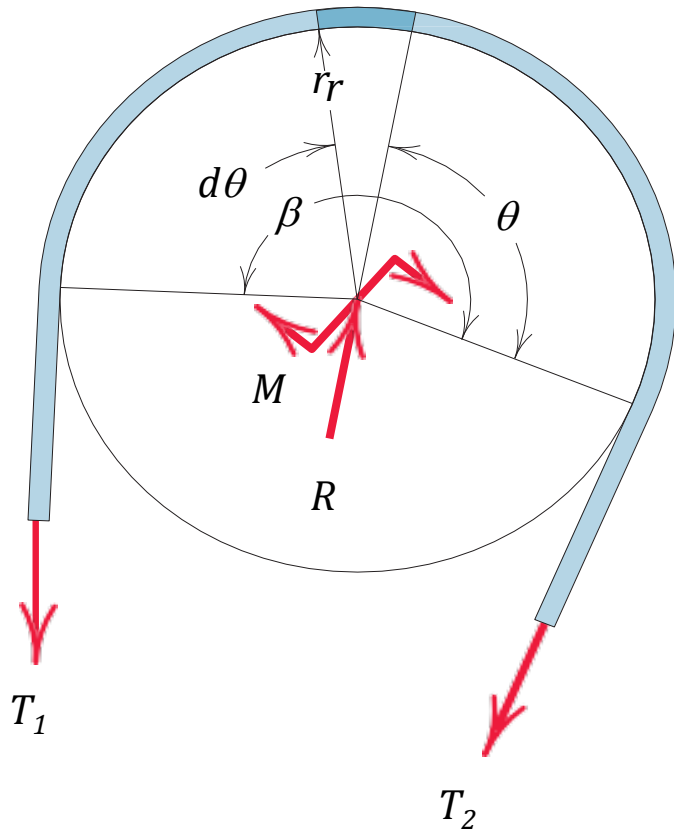
$$3750 \cos 51.6^\circ - P \cos 21.7^\circ = 0$$

$$P = 2500 \text{ N}$$

# Graphical Solution

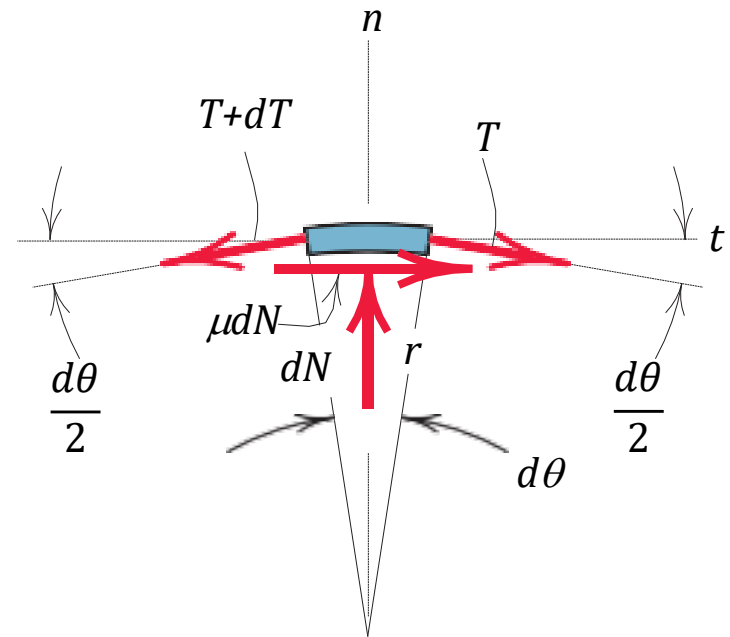
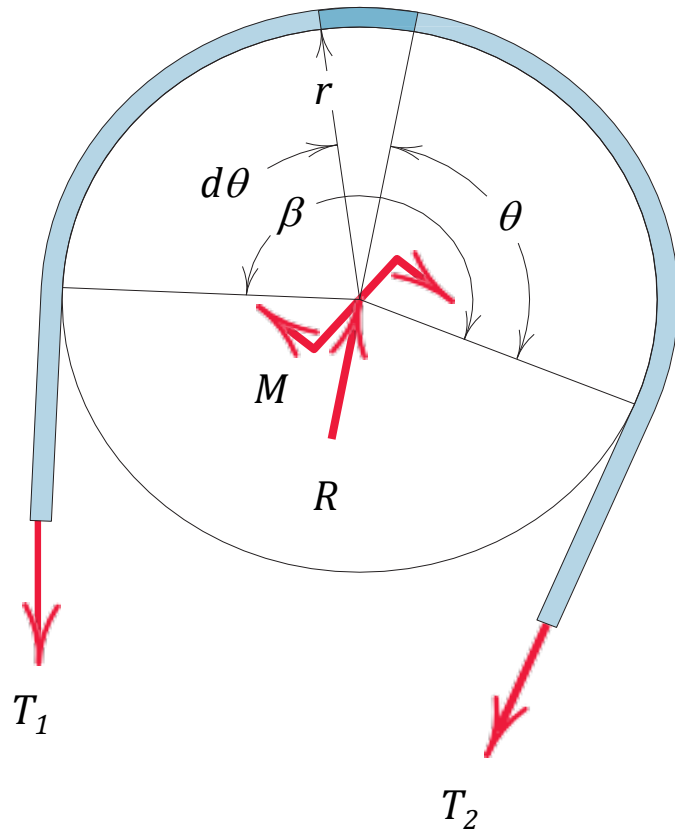


# FLEXIBLE BELTS



- The impending slippage of flexible cables, belts, and ropes over sheaves and drums is important in the design of belt drives of all types, band brakes, and hoisting rigs.
- Drum is subjected to the two belt tensions  $T_1$  and  $T_2$ , the torque  $M$  necessary to prevent rotation, and a bearing reaction  $R$ .
- With  $M$  in the direction shown,  $T_2$  is greater than  $T_1$ .

# FLEXIBLE BELTS



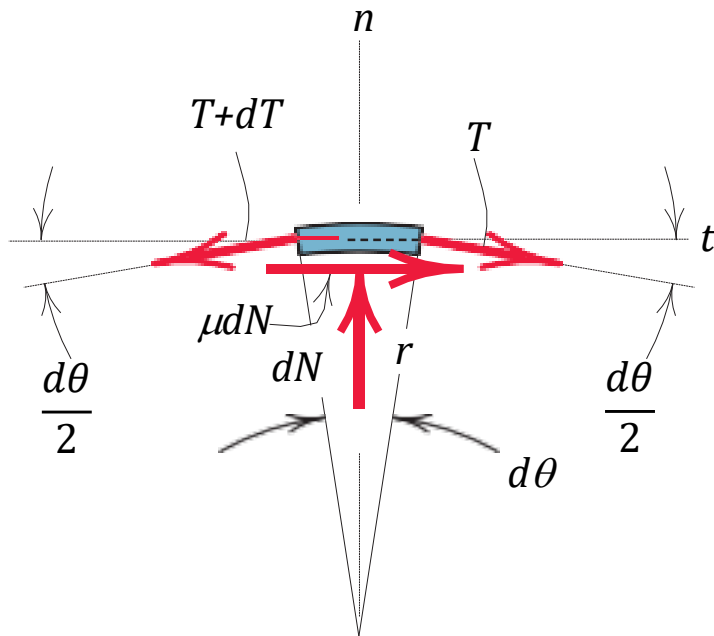
# FLEXIBLE BELTS

t:  $T \cos \frac{d\theta}{2} + \mu dN = (T + dT) \cos \frac{d\theta}{2}$

$\mu dN = dT$   $\leftarrow \cos \frac{d\theta}{2} = 1$

n:  $dN = (T + dT) \sin \frac{d\theta}{2} + T \sin \frac{d\theta}{2}$

$dN = T d\theta$   $\leftarrow \sin \frac{d\theta}{2} = \frac{d\theta}{2}$



Combining both

$$\frac{dT}{T} = \mu d\theta$$

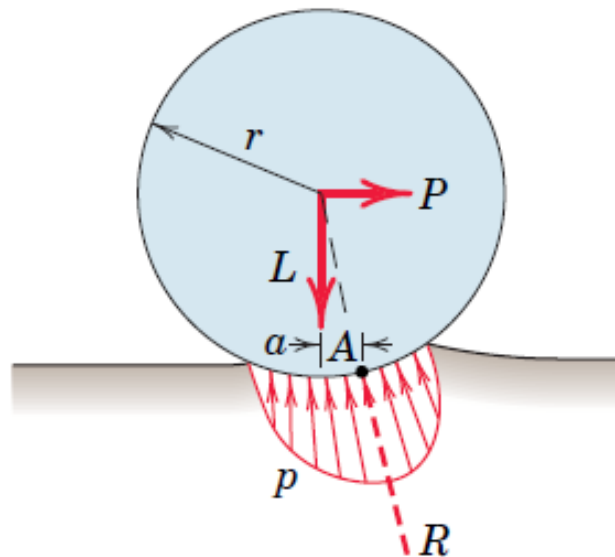
$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^{\beta} \mu d\theta$$

$$\ln \frac{T_2}{T_1} = \mu \beta$$

$$T_2 = T_1 e^{\mu \beta}$$

## ROLLING RESISTANCE

- Deformation at the point of contact between a rolling wheel and its supporting surface introduces a resistance to rolling,
- This resistance is not due to tangential friction forces
- Therefore is an entirely different phenomenon from that of dry friction.

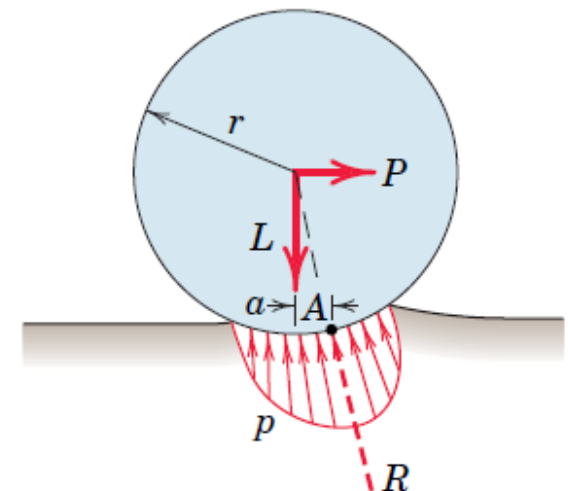


## ROLLING RESISTANCE

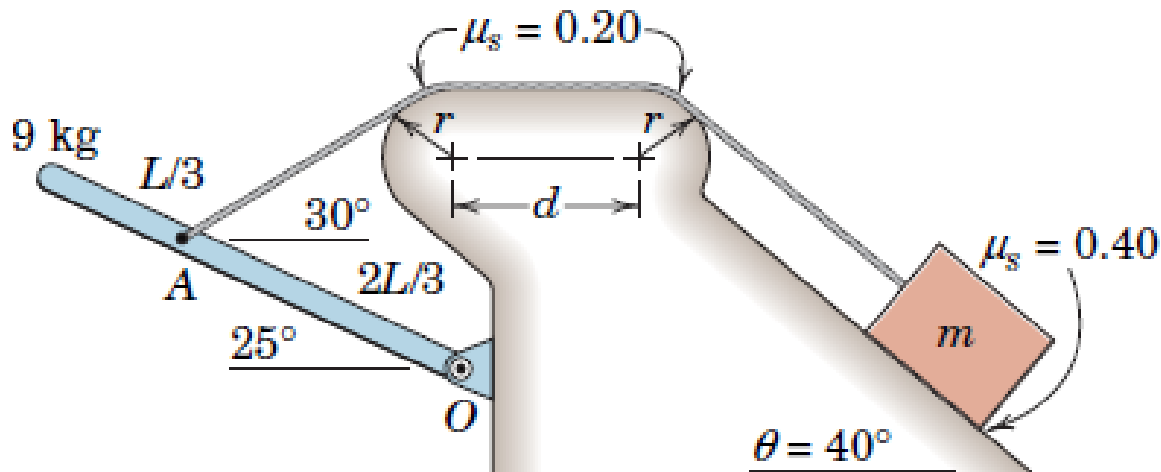
- The **resultant**  $R$  of this distribution acts at some point  $A$  and must **pass through the wheel center** for the wheel to be in equilibrium.
- We find the **force**  $P$  necessary to maintain rolling at constant speed by equating the **moments of all forces about  $A$  to zero**.
- This gives us

$$P = \frac{a}{r} L = \mu_r L$$

$$\mu_r = \frac{a}{r}$$



- Example 1: Determine the range of mass  $m$  over which the system is in static equilibrium. The coefficient of static friction between the cord and the upper curved surface is 0.20, while that between the block and the incline is 0.40. Neglect friction at the pivot  $O$ .



# Motion of m impends up the incline.

$$[\Sigma M_O = 0]$$

$$-T_A \left( \frac{2L}{3} \cos 35^\circ \right) + 9(9.81) \left( \frac{L}{2} \cos 25^\circ \right) = 0$$

$$T_A = 73.3 \text{ N} = T_2$$

$$[\Sigma F_y = 0]$$

$$N - mg \cos 40^\circ = 0$$

$$N = 0.766mg$$

$$[T_2 = T_1 e^{\mu_s \beta}]$$

$$73.3 = T_1 e^{0.20[30^\circ + 40^\circ]\pi/180^\circ}$$

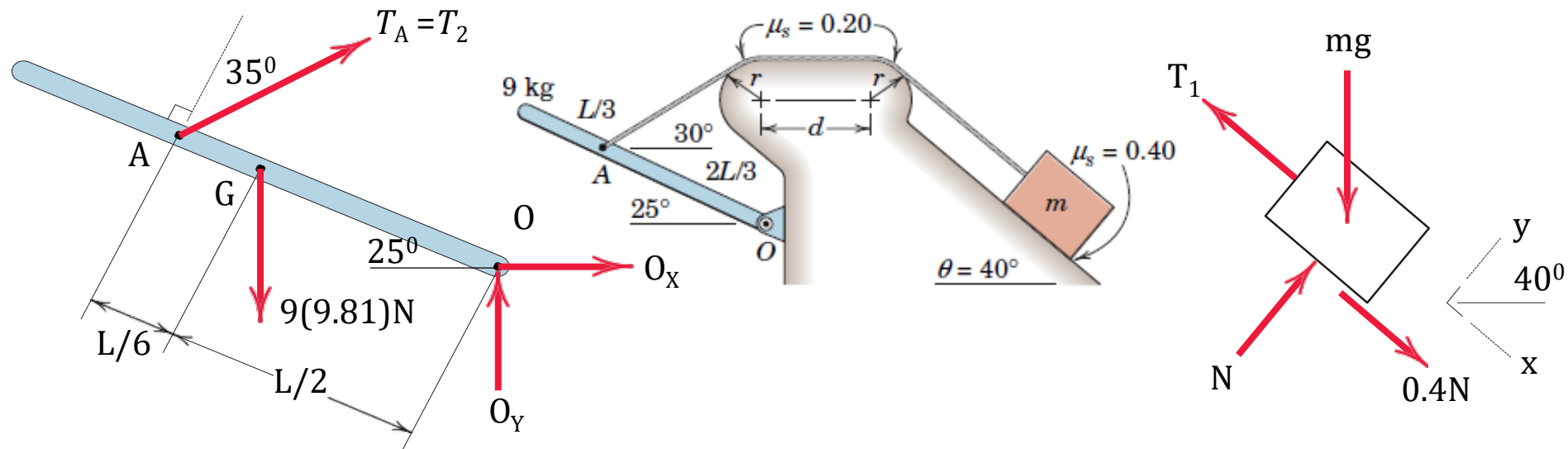
$$[\Sigma F_x = 0]$$

$$-57.4 + mg \sin 40^\circ + 0.40(0.766mg) = 0$$

$$T_1 = 57.4 \text{ N}$$

$$mg = 60.5 \text{ N}$$

$$m = 6.16 \text{ kg}$$



# Motion of m impends down the incline

$$[\Sigma M_O = 0]$$

$$-T_A \left( \frac{2L}{3} \cos 35^\circ \right) + 9(9.81) \left( \frac{L}{2} \cos 25^\circ \right) = 0$$

$$T_A = 73.3 \text{ N} = T_1$$

$$[T_2 = T_1 e^{\mu_s \beta}]$$

$$T_2 = 73.3 e^{0.20[30^\circ + 40^\circ]\pi/180^\circ}$$

$$T_2 = 93.5 \text{ N}$$

$$[\Sigma F_y = 0]$$

$$N - mg \cos 40^\circ = 0$$

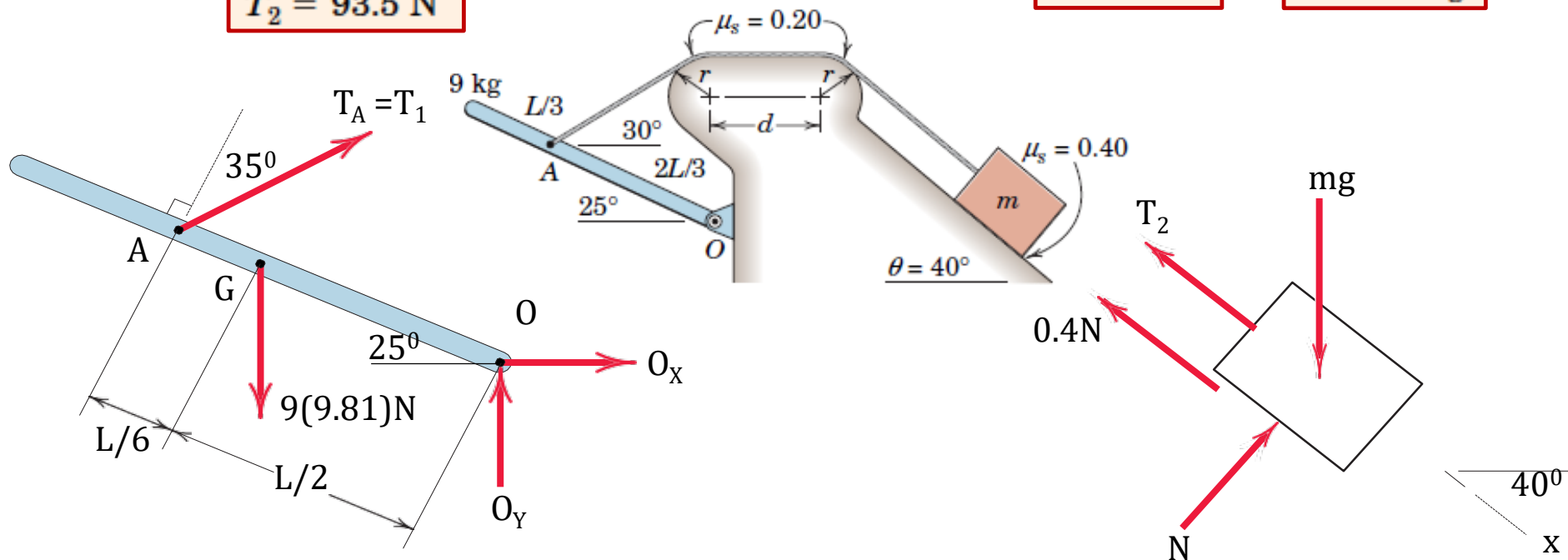
$$N = 0.766mg$$

$$[\Sigma F_x = 0]$$

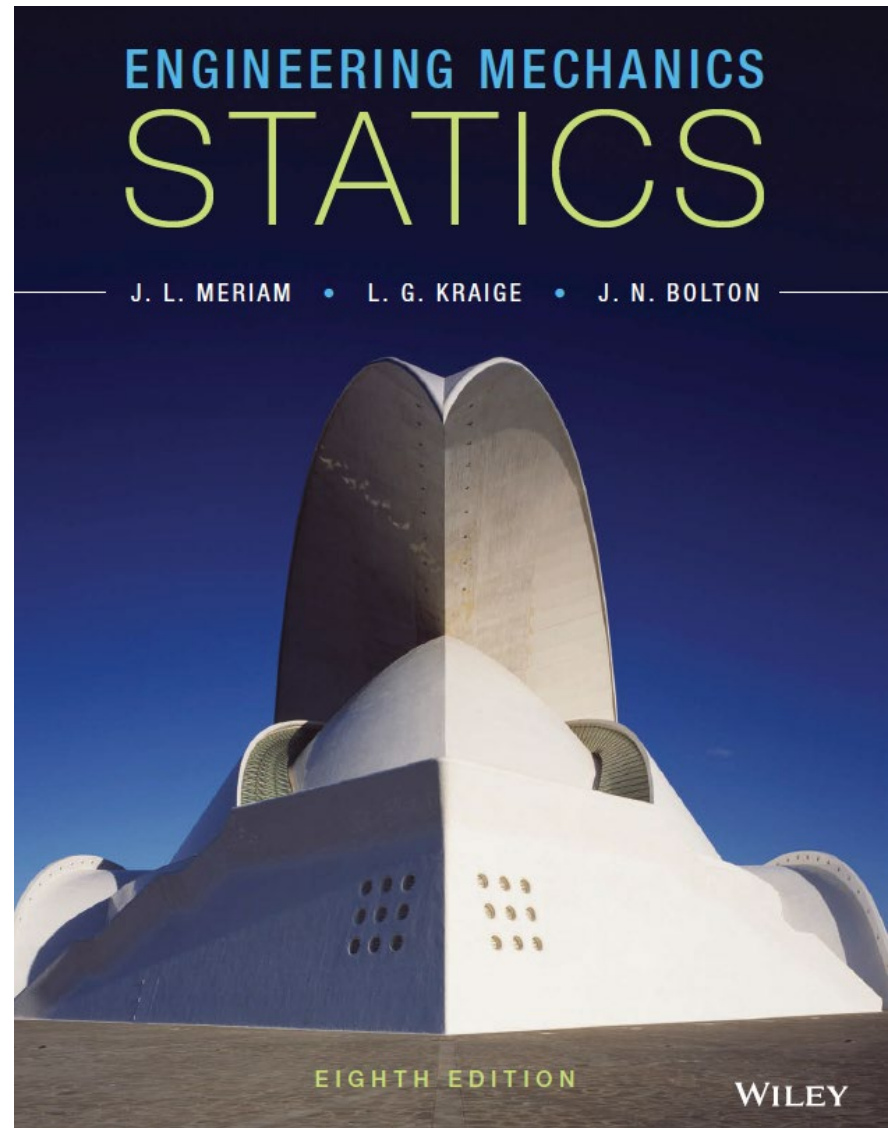
$$-93.5 - 0.4(0.766mg) + mg \sin 40^\circ = 0$$

$$mg = 278 \text{ N}$$

$$m = 28.3 \text{ kg}$$



## References :



# THANK YOU